Constraints on the cosmic-ray electron spectrum above 25 TeV from the LHAASO experiment

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Cosmic-ray electrons and positrons (CREs) offer valuable insights into the origins of local cosmic rays, as they rapidly cool in interstellar magnetic and radiation fields, limiting their propagation range. The CRE spectrum reveals a spectral break around 1 TeV, followed by a continuous power-law behavior that possibly extends beyond 40 TeV. Deviations from a pure power-law in this spectrum could provide important clues about the local origins of CREs. However, due to the rapidly falling flux of CRE, the ratio of hadron/CRE increases with energy, making the rejection more and more challenging. Equipped with a large detection aperture and strong background rejection capability, LHAASO has the potential to extend CRE spectrum measurements beyond 25 TeV, which is almost the highest energy measured by existing experiments. In this work, we study the detector performance and explore the optimized background rejection of cosmic nuclei background specifically for CRE observations in LHAASO. Our analysis reveals that muon-poor nuclei showers have a strong dependence on the hadronic model when estimating the cosmic nuclei background and rejection capability. Consequently, we derived a 90% CL upper limit for the CRE spectrum from 25 to 160 TeV, providing a new and restrictive constraint in the hundreds of TeV range for the first time.

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I. INTRODUCTION

Cosmic rays are high-energy particles originating from outer space, primarily composed of protons, helium nuclei, and heavier nuclei. A small fraction component, referred to

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as cosmic-ray electrons (CREs), includes cosmic-ray electrons and positrons (Hereafter, CREs are generally referred to as electron and positron). In contrast to cosmic nuclei, CREs have limited propagation distances due to energy losses from synchrotron radiation and inverse Compton scattering, which confine TeV CREs approximately within 1 kpc of local region over a timescale of 10^5 yrs [1] and references therein [2–4]. Consequently, the spatial distribution of nearby sources and propagation models can significantly affect observed spectral features in CREs at the highest energies [5–7]. Measuring the CRE spectrum beyond TeV energies is therefore crucial for probing nearby astrophysical sources, cosmic-ray propagation, and acceleration mechanisms.

Several space-based experiments have advanced our understanding of CREs. Spectrometers like AMS-02 [8,9], BESS [10], HEAT-pbar [11], and PAMELA [12,13] have provided insights into individual electron and positron spectra, while calorimeters like Fermi-LAT [14], CALET [15], and DAMPE [16] have extended the energy range up to 4 TeV for CRE measurements. Observed suppression of the CRE spectrum around TeV energies may be tied to the discrete distribution of local sources [17] or energy-dependent diffusion [18]. Additionally, positron measurements indicate a deviation from secondary production models [19,20], suggesting possible astrophysical origins [21] or new physics, such as dark matter annihilation [22]. Space-based instruments, however, face challenges in measuring CREs at high energies due to limited acceptance and softened spectra above tens of TeV energy range, resulting in reduced signal statistics.

Ground-based experiments like Imaging Atmospheric Cherenkov Telescopes (IACTs) such as H.E.S.S. [23], MAGIC [24], and VERITAS [25] have furthered CRE observations up to around 5 TeV, with H.E.S.S. observing the softening of the spectrum around 1 TeV and a featureless power-law potentially extending to 40 TeV [26–28]. If nearby sources significantly contribute to CREs, features like a cutoff in the source spectrum or a cooling break may cause deviations from a simple power-law behavior. This motivates the extension of CRE observations beyond 25 TeV to explore their origins. Yet, IACTs are constrained by energy coverage and aperture limitations in this range.

The Large High Altitude Air Shower Observatory (LHAASO), as a ground-based experiment with a large collection area, wide field-of-view, high cosmic-ray rejection, and nearly continuous observation capability, presents a unique opportunity to study CREs. Section II describes the experimental setup and two challenges background when dealing with CRE measurements. Section III gives detailed the results of KM2A's performance on CRE observation, such as energy reconstruction, effective aperture, and lepton/hadronic separation. The method of derivation on upper limits of CRE fluxes are introduced in Sec. IV. The astrophysical implication and systematic uncertainty are discussed in Sec. V.

II. THE LHAASO EXPERIMENT

The LHAASO is a ground-based extensive air shower (EAS) detector array located 4,410 m above sea level [29]. It employs a hybrid detection technique and consists of three subarrays—Square Kilometer array (KM2A), Water Cherenkov Detector Array (WCDA), and Wide Field Cherenkov Telescope Array (WFCTA). Detailed information on the configuration of LHAASO can be found in [30]. KM2A is made up of 5216 electromagnetic particle detectors (ED) and 1188 muon detectors (MD), covering an energy range of 30 TeV to 10 PeV. It is capable of measuring secondary electromagnetic particles and muons in air showers using EDs and MDs respectively, which can be used for composition discrimination. Previous results from the KM2A in observing the Crab Nebula demonstrate robust background rejection capabilities, achieve over 2×10^4 at the energies above 100 TeV [31], particularly advantageous for electron spectrum measurements. WCDA consists of 3,120 homogeneous water Cherenkov detectors, covering an area of $78,000\ \mathrm{m}^2$ at the center of LHAASO. The WCDA covers an energy range from 100 GeV to 30 TeV and achieves a rejection power of 500 at 6 TeV [32], which is approximately an order of magnitude below the requirement to observe CREs signal. This study focuses on energies above 25 TeV, thus only KM2A is utilized.

There are two types of background that affect the measurement of the electron signal. The majority of events detected by KM2A originate from hadronic cosmic rays, primarily protons and nuclei, whose fluxes have been directly measured by space-based detectors such as AMS-02 [33], DAMPE [34,35], and the balloon-borne experiment CREAM [36,37]. These measurements span from tens of GeV to several hundred TeV, with overlapping energy ranges that provide consistent coverage of the nuclei spectrum. Based on the extrapolation of the electron flux as reported by H.E.S.S. [28], the electron flux constitutes approximately 10^{-5} of the nuclei flux at 20 TeV, decreasing to 10^{-6} above 200 TeV. This background can be strongly suppressed by using electron and hadron discrimination, but it still poses significant challenges. The other kind of background is gamma-ray showers. Both gamma rays and electrons initiate electromagnetic cascades in the atmosphere. It is almost impossible to distinguish electron-induced showers from gamma-ray showers using the ground-based sampling EAS detector. Another issue is that the presence of the extragalactic gamma-ray background (EGB) cannot be rejected and has to be accepted as part of the CRE signal events. Fortunately, Fermi-LAT has measured the EGB from 100 MeV to 820 GeV and found it to be two orders of magnitude below the CRE measurement [38]. The spectrum follows a power-law behavior with an exponential cutoff at

230 GeV, making the contamination of EGB negligible in the TeV regime.

III. KM2A'S PERFORMANCE IN CRE OBSERVATIONS

KM2A's performance in gamma-ray-induced showers has been studied through observations of the Crab Nebula [31]. For electron-induced showers, simulations are used due to the lack of a standard candle. The EAS processes of electrons and hadrons are both simulated using the CORSIKA code (v77410) [39]. The attenuation length of the muon content has been previously compared by LHAASO using three high-energy hadronic interaction models (QGSJET-II-04 [40,41], SIBYLL 2.3d [42], and EPOS-LHC [43]). Among them, EPOS shows better agreement with the experimental results, while QGSJET-II-04 and SIBYLL 2.3d tend to predict a longer muon attenuation length, corresponding to more muons in the shower development [44]. Thus, the EPOS model was used for the electron upper limit, and QGSJET-II-04 was employed to estimate systematic uncertainties. For hadronic interactions below 80 GeV, the FLUKA code [45] was utilized, while electromagnetic interactions were simulated with the EGS4 framework [46,47]. To accurately simulate the response of the KM2A detector, the G4KM2A software was developed based on the Geant4 package (v4.10.00) [48]. The validity of G4KM2A has been verified through observations of the Crab Nebula.

Electron-induced air showers and nuclei-induced showers are simulated over a wide energy range from 10^{12} to 10^{15} eV. The showers follow an E^{-2} energy spectrum and an isotropic angular distribution, with a zenith angle range of 0°–40° and an azimuth range of 0°–360°. The shower cores of the injected showers are uniformly distributed in a ring-shaped area from 200 to 480 m from the array center. This region is chosen to optimize KM2A's performance, as it avoids areas where muon detectors are absent. As a result, the total number of simulated events for CREs is 2.2×10^8, while the total number of simulated events for five groups of cosmic nuclei (proton, helium, nitrogen, aluminum, and iron) is approximately 5×10^8. The simulated CRE events were weighted according to the spectrum reported by the H.E.S.S. experiment [27], while cosmic-ray events were weighted based on various composition models, including the Gaisser H3a [49] and Hörandel [50] models. In our analysis of experimental data, we excluded contributions from both the Galactic plane and individual gamma-ray sources that are significantly detected by KM2A, including those located at high Galactic latitudes. To ensure consistency with these experimental data selection criteria, we assigned each simulated event a pseudotimestamp, allowing us to obtain the Galactic coordinates accordingly.

We imposed stricter selection criteria on the position and zenith angle of the showers compared to those used in previous gamma-ray observations and requiring reconstructed shower cores to be located within a ring-shaped area of 320–420 m from the center of LHAASO and restricting the zenith angle to less than 30°. In this way an erroneous reconstruction of showers with cores outside the array can be mostly avoided, and all muons and electromagnetic particles counted in the distance range 40–200 m from the shower axis with the full KM2A EDs + MDs is guaranteed. Due to the computational challenges of simulating enough high-energy zero-muon events and the rapid decline of the CREs with energy, we focus on the 25–160 TeV range.

A. Energy reconstruction

For the event reconstruction, the shower core and direction reconstruction algorithms follow the same rules as those used for gamma-ray showers. As for the energy reconstruction of gamma-ray induced shower [31], the particle density at a radial distance of 50 m from the shower core, denoted as ρ_{50}, is employed to estimate the energy of CREs. Using electron shower simulation data, we optimized the parameters based on the relationship between ρ_{50} and the true primary energy, thus refining the energy estimation for electron-induced showers.

As shown in Fig. 1, the left panel illustrates that the energy resolution function does not follow a normal distribution; consequently, the energy resolution is defined as $(E_{\rm rec} - E_{\rm true})/E_{\rm true}$ including 68% events for each reconstructed energy bin, while the bias is defined as the mean of the resolution function. The energy resolution is approximately 21% at 25 TeV, improving to 12% at 100 TeV, while keeping minimal bias as presented in Fig. 1 right panel.

B. Lepton/hadron discrimination

For lepton/hadron separation in gamma-ray detection [31], the KM2A array uses the muon-to-electromagnetic-particle ratio, defined as $R = \log[(N_\mu + 10^{-4})/N_e]$. This analysis employs the same variable for studying electron-induced showers, and R is good for rejecting cosmic-ray events, as Fig. 2 shows the comparison of the distribution of the parameter R between the cosmic-ray background simulation and the observational data. In the energy range of 25 to 160 TeV, there are about 54.3% electron-induced showers that are muon-poor events corresponds to the population of $R < -6$ ($N_\mu = 0, N_e > 100$), whereas only approximately 3.0×10^{-4} observed events are muon-poor. Based on simulations, electron-induced events are expected to constitute approximately 1.1×10^{-5} of the total observed events.

The comparison of R distributions suggests that EPOS–FLUKA achieves good agreement with data in the R distribution around the peak region ($-2 < R < -0.5$), with the MC-to-data ratio varying between −5.2% and +8.2%. Similarly, QGSJET–FLUKA provides a close match within

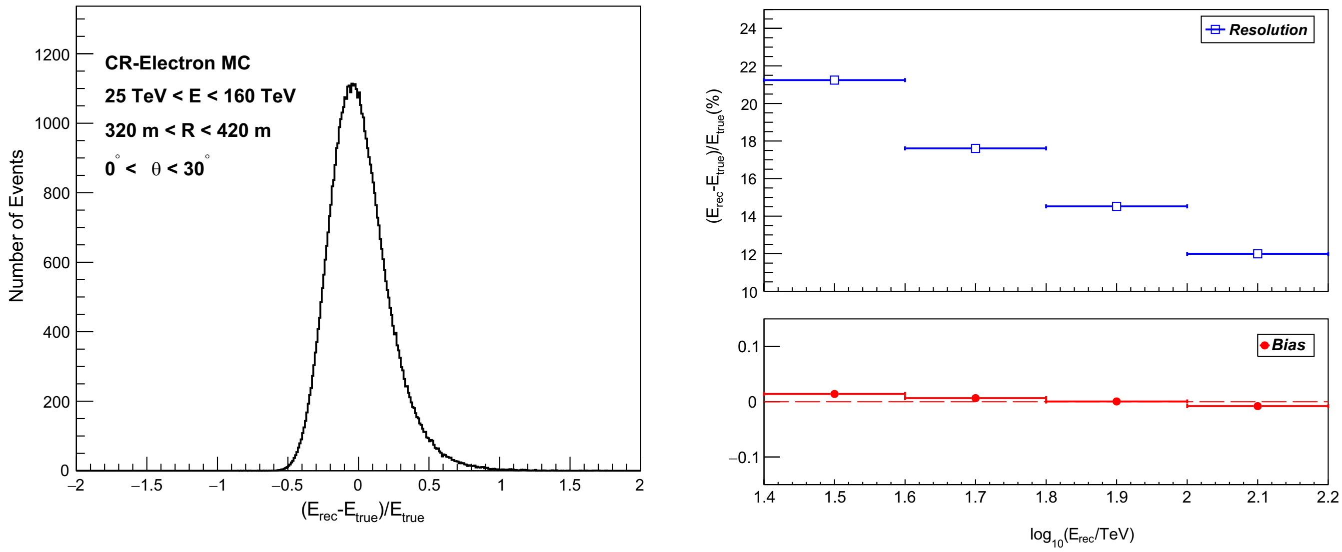

FIG. 1. The electron-induced shower simulation data: the left panel shows the energy resolution function of showers in the energy range of 25–160 TeV with zenith angle 0°–30°; the right panel shows energy resolution (top, blue squares) and bias (bottom, red filled circles) of the reconstructed energy.

$-1.5 < R < -0.5$, with the MC-to-data ratio varying between -15.1% and $+0.4\%$. However, it shows a systematic deviation of approximately -20% compared with the data within $-2 < R < -1.5$. While both hadronic interaction models show reasonable agreement with the data in the peak region, suggesting that most events are well predicted, notable discrepancies arise for events with a low number of muons ($R < -2$). Specifically, in the muon-poor region, the EPOS–FLUKA MC-to-data ratio rises to around 2, whereas the QGSJET–FLUKA MC-to-data ratio drops to approximately 0.2. This discrepancy introduces uncertainties in background estimation due to secondary muons predicted by different hadronic interaction models.

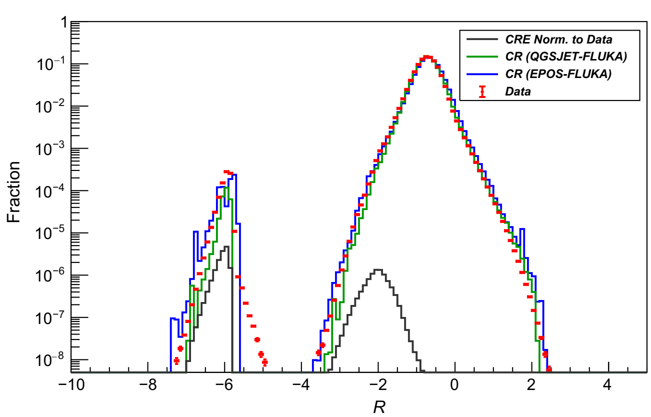

FIG. 2. The comparison of lepton/hadron separation parameter, R, between the cosmic-ray background simulation with observation data applied in the same data quality criteria after MC/Data comparison from 25 to 160 TeV. The distribution of R from simulated electron events is normalized to the observation data to illustrate the expected relative abundance.

In general, the parameter R is optimized for a typical pointlike source analysis based on simulations, such as Crab [31]. In this work, we restricted the criteria to further suppress the cosmic-ray background at high energies, thus obtaining a maximum detection significance, as followed by Eq. (1) in Sec. IV, ensuring that the survival fraction of electron-induced events (ϵ_{CRE}) remains above 30%. Figure 3 illustrates the survival fraction of electron-induced showers and cosmic-ray background events (from both simulation and observation) after applying the discrimination criteria (R_{opt}) optimized from the cosmic-ray Monte Carlo sample. The optimal parameter set is $R_{\mathrm{opt}} < -6.10$ at 25 TeV, becoming more stringent with increasing energy, reaching $R_{\mathrm{opt}} < -6.45$ at 100 TeV. For energies exceeding 100 TeV, the

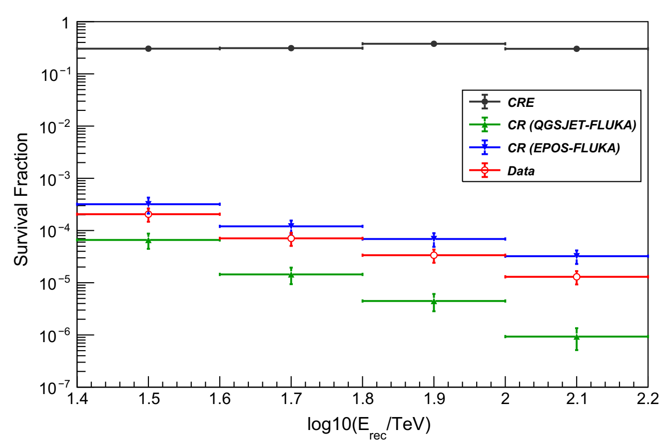

FIG. 3. The survival fraction of simulated CRE (ϵ_{CRE}), cosmic-ray background ($\epsilon_{\mathrm{CR,models}}$), and observed events ($\epsilon_{\mathrm{Data}}$) in different energy bins after the discrimination criteria optimized from the CR MC sample.

discrimination criterion is relaxed to $R_{\mathrm{opt}} < -3.25$ due to the low statistics for muon-poor CR events. The survival fraction for gamma-ray showers remains at 30.3%, and the rejection power for cosmic-ray-induced showers exceeds 3.1×10^4 at energies greater than 100 TeV.

In particular, the EPOS-FLUKA model shows better consistency with the observational data compared to the QGSJET-FLUKA model in terms of survival fraction. Under the EPOS-FLUKA model, the ratio of cosmic-ray survival rate to the data survival rate increases with energy, from 1.55 to 2.48. In contrast, under the QGSJET-FLUKA model, this ratio decreases with increasing energy, from 0.32 to 0.07, showing a difference of more than an order of magnitude. The QGSJET-FLUKA model predicts a larger discrepancy with observational results, leading to model dependence in background estimation. This result is also supported by another KM2A's study on the muon content in air showers [44], where the muon attenuation length serves as a sensitive indicator for hadronic interaction models. The attenuation lengths of muon content predicted by QGSJET-FLUKA were longer than those observed in LHAASO data, whereas the results from EPOS-FLUKA were relatively closer to the LHAASO data.

To quantify the influence of different models, the number of muon-poor events serves as an intrinsic indicator of background estimation, independent of the discrimination criteria. In the energy range of 100–160 TeV, the relative difference due to hadronic interaction models is approximately a factor of 2, while the relative difference between composition models is around 16.0%. Compared to composition models, hadronic interaction models introduce substantial uncertainty in determining the electron signal excess. However, based on our observations, it is not possible to favor one hadronic interaction model over another, particularly since the discrepancies are concentrated in a minority of events (i.e., muon-poor events).

C. Effective aperture

To calculate the flux of the CREs, the effective aperture, A_{eff}, is determined to estimate the detection efficiency. The effective aperture reaches a constant value, $135,000\ \mathrm{m}^2 \cdot \mathrm{sr}$, at energies greater than 25 TeV with applying the data quality criteria only. The implementation of lepton/hadron separation with R_{opt} reduces detection efficiency, thus affecting the aperture used for the electron flux analysis later, as shown in Fig. 4. In this work, the aperture reaches about $54{,}700\ \mathrm{m}^2 \cdot \mathrm{sr}$ at 25 TeV and achieves about $42{,}100\ \mathrm{m}^2 \cdot \mathrm{sr}$ over 100 TeV with more relaxed criteria on R_{opt}. The aperture declines around 40 TeV when applying lepton/hadron discrimination, corresponding to a drop in the survival fraction of electrons at the same energy. The uncertainties considered include statistical errors and reweighting-induced errors. Within the energy range of 25 to 160 TeV, the uncertainty increases from 0.22% to 0.51%. The uncertainty of effective aperture is too small to be distinctly visible on a logarithmic scale.

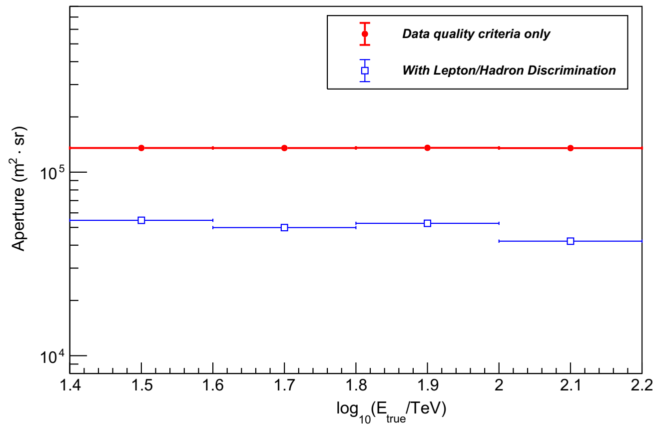

FIG. 4. The aperture of the KM2A for electron showers with data quality criteria applied. Red dots represent data before discrimination, while blue squares indicate data after lepton and hadron discrimination.

IV. ANALYSIS AND RESULTS

A. Measurement data

The present study utilizes measurement data collected by the full KM2A array from July 21, 2021 to July 31, 2023, spanning a total of 737 days. During this period, KM2A array recorded approximately 1.86×10^{10} events. Due to the numerous gamma-ray sources and the diffuse gamma-ray background observed by KM2A in the Galactic plane [51,52], we exclude the region within $\pm 15°$ to remove the gamma background from the galactic plane. Additionally, four significant sources remain at high Galactic latitudes ($|b| > 15°$) in the KM2A data for $E > 25$ TeV, as identified in the first LHAASO catalog [51]. These sources are masked following the method described in [52]. Assuming that the angular distribution of cosmic-ray electrons in the relevant energy range is nearly isotropic, the Galactic latitude cut reduced the field of view by 25.40%, while the masking of gamma-ray sources further reduced it by 0.05%, leading to a total reduction of 25.45%.

B. Upper limits of CRE fluxes

Although KM2A provides good energy estimates for CREs and a sufficiently large effective aperture to gather ample statistics on CRE events, the primary source of uncertainty comes from different hadronic models. These models predict different cosmic-ray background levels, which introduce significant uncertainties. With better consistency compared to observed data, the EPOS-FLUKA interaction model, normalized to the Hörandel composition model, provides a moderate estimation of the cosmic-ray background after applying lepton/hadron discrimination. Based on the optimized discrimination criteria, the significance S, could be formulated as follows,

TABLE I. KM2A performance summary: Energy bins, energy resolution, electron and hadron survival probabilities, observed event numbers, expected hadron numbers, and electron flux upper limits from the EPOS model.

E (TeV)	E resolution	ϵ_{CRE}	$\epsilon_{\mathrm{QGSJET}}$	ϵ_{EPOS}	N_{obs}	$N_{\mathrm{exp,EPOS}}$	ξ_{limit}	$E^3J(E)$ $(\mathrm{GeV^2\ m^{-2}\ sr^{-1}\ s^{-1}})$
25–40	21.24%	0.30	6.59×10^{-5}	3.19×10^{-4}	3.43×10^{5}	3.28×10^{5}	1.58×10^{5}	1.68×10^{2}
40–63	17.61%	0.31	1.44×10^{-5}	1.20×10^{-4}	5.24×10^{4}	6.18×10^{4}	1.67×10^{4}	4.03×10^{1}
63–100	14.52%	0.38	4.47×10^{-6}	6.88×10^{-5}	9.30×10^{3}	1.78×10^{4}	2.93×10^{3}	1.37×10^{1}
100–160	11.99%	0.30	9.30×10^{-7}	3.22×10^{-5}	1.12×10^{3}	3.93×10^{3}	4.89×10^{2}	6.99×10^{0}

$$S = \frac{N_{\mathrm{sig}}}{\sigma(N_{\mathrm{tot}})} = \frac{N_{\mathrm{CRE}}}{\sqrt{N_{\mathrm{CR}} + N_{\mathrm{CRE}} + (\delta \cdot N_{\mathrm{CR}})^2}}, \qquad (1)$$

where N_{sig} and N_{tot} represent the number of the expected signal and total events after discrimination, N_{CRE} and N_{CR} represent the number of simulated electron and cosmic-ray background events after discrimination, respectively. An additional term representing the uncertainty in the absolute flux normalization ($\delta = 20\%$) is included based on the cross-calibration of the energy scales from different ground-based experiments [53]. This level of uncertainty is significant and could potentially overwhelm the surviving CRE signals.

Beyond this, the uncertainty introduced by the hadronic interaction models is even larger than the uncertainty in absolute flux normalization, further complicating the significant detection of electron events. The observed significance is less than 3, based on the EPOS-FLUKA interaction model, which reflects the insufficient rejection power compared to expectations as shown in Fig. 3. Given this, it is not feasible to derive a CRE flux beyond 25 TeV. Instead, a conservative upper limit on the CRE flux can be estimated using a hypothesis testing approach based on both observed events and simulated backgrounds. Since cosmic-ray background events dominate, there are ample surviving events in each energy bin. We treat the surviving cosmic-ray background and observed events after discrimination as two independent Gaussian-like distributions. As a result, the number of excess events, ξ, expected to be from CREs, approximately follows a normal distribution, $\xi \sim \mathcal{N}(\mu, \sigma^2)$. And the expected mean (μ) and variance (σ^2) can be expressed as follows:

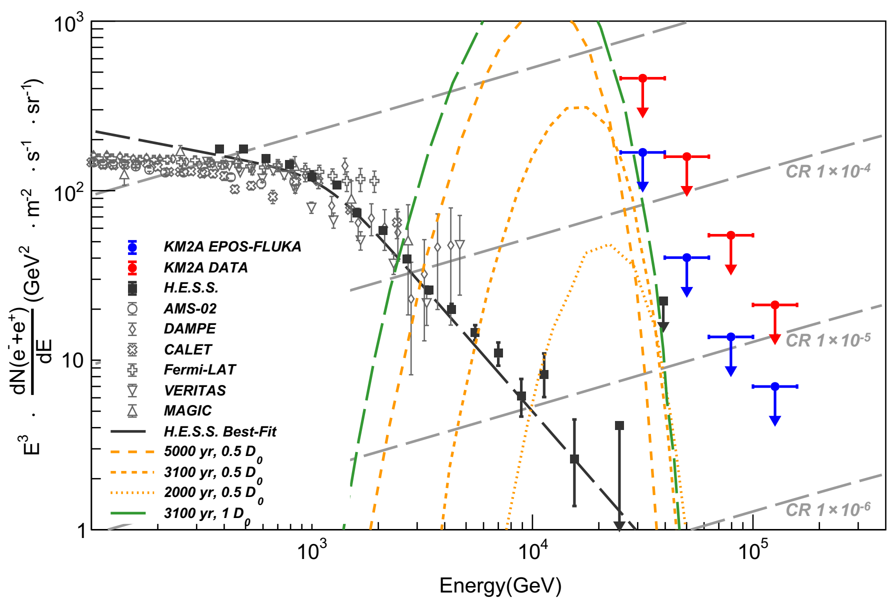

FIG. 5. The KM2A 90% CL upper limit for the CRE spectrum. The CRE spectrum results from various experiments are also shown, including AMS-02 [8,9], Fermi-LAT [14], DAMPE [14], CALET [15], H.E.S.S. [28], VERITAS [25], and MAGIC [24]. The black dash-dot line represents the best-fit of the all-electron spectrum measured by H.E.S.S in 2024 [28]. KM2A's constraints on the parameter space for potential nearby astrophysical sources of cosmic-ray electrons. The predicted CRE spectrum is assumed to be dominantly contributed by a local PWN, such as Vela-X. A Vela-type sources is modeled using different diffusion coefficients at 10 GeV, $D_0 = 10^{28}\ \mathrm{cm^2/s}$, (green long-dashed) and injection times: 5000 yrs (orange medium-dashed), 3100 yrs (orange short-dashed), and 2000 yrs (orange dotted), as described in [56]. The injection age and diffusion coefficient are treated as free parameters to explore the simplified model templates, demonstrating KM2A's constraint capability.

$$\mu = N_{\text{obs}} - N_{\text{CR}},$$
$$\sigma^2 = \sigma^2_{N_{\text{obs}}} + \sigma^2_{N_{\text{CR}}} + (\delta \cdot N_{\text{CR}})^2, \tag{2}$$

where N_{obs} is the number of observed events after discrimination. To calculate an upper limit on the CRE flux, a one-sided confidence interval is used. It is important to exclude nonphysical cases where the excess electron signal (ξ) is negative from the analysis. Thus, we incorporated prior knowledge of the background into our calculations, following established methods [54,55]. The upper limit of the electron signal ξ_{limit} is determined using the complementary cumulative distribution function (CCDF) of the normal distribution as follows:

$$\alpha = \frac{\Phi(\xi_{\text{limit}})}{\Phi(0)}, \tag{3}$$

where $1-\alpha$ denotes the confidence level, and $\Phi(\xi)$ represents the CCDF. When $\alpha = 0.1$, the upper limit on the excess electron signal with a 90% confidence level, ξ_{limit}, is determined. The ability of KM2A to constrain the upper limit on CRE flux is then given by

$$J(E) = \frac{\xi_{\text{limit}}}{\Delta E \cdot A_{\text{eff}} \cdot T}, \tag{4}$$

where ξ_{limit} represents the excess signal in each energy bin, ΔE is the energy interval, T is the observation time, and A_{eff} is the effective aperture. All key data relevant to the upper limit of the electron spectrum in this data analysis section are summarized in Table I. The 90% confidence level upper limit on the CRE flux, based on KM2A observations, is shown in Fig. 5, along with measurements from satellite and ground-based experiments. The upper limit shown in blue represents results predicted based on the model-estimated cosmic-ray background. KM2A's upper limit exceeds the best fit of H.E.S.S. results on best-fit CRE spectrum but approaches to it as energy increases. Although the upper limit established by KM2A is approximately twice as high as the results reported by H.E.S.S. in 2024 at 40 TeV, KM2A provides an upper limit on the CRE spectrum, approximately 5.0×10^{-6} of the all-CR spectrum, to constrain a potential upturn in the energy range above 100 TeV. The upper limit shown in red represents results derived from the surviving observed events, treated as electron signal candidates without removing cosmic-ray background contamination, which provides a more conservative constraint on the upper limit of the CRE spectrum.

V. DISCUSSION

In the astrophysical source scenario, the shape of the cosmic-ray electron spectrum in the TeV energy range is generally attributed to contributions from young, nearby sources such as pulsar, pulsar wind nebulae (PWNe) or supernova remnants (SNRs). Vela is identified as a candidate local source of CRE at TeV energy due to its suitable age, distance, and strong observed radio flux [1,57]. Within the Vela region, Vela X, powered by the pulsar, PSR B0833–45, is one of the most well-studied PWNe across multiple wavelengths [58–62]. Hence, Vela X can serve as a useful template for comparing the CRE spectrum upper limits derived from this study, providing constraints on the parameter space of astrophysical sources. Assuming a Vela-type source modeled by a simple burstlike injection into the interstellar medium and its electron spectrum adapted to match power-law behavior with an exponential cutoff, the injection age and diffusion coefficient are treated as free parameters. Meanwhile, the injection energy, spectral index, energy cutoff, and magnetic field are derived through observations, as described in [56]. As shown in Fig. 5, the orange dotted and dashed lines illustrate the effects of varying injection ages, while the green dash dotted line represents the impact of different diffusion coefficients. It can be seen that shorter injection times restrict both the energy range and flux intensity, while slightly enhancing the flux in the highest-energy region. On the other hand, a lower diffusion coefficient means a slower propagation and less electron with lower energy are able to arrive at Earth, hence it results in a more gradual decline in flux, with higher flux sustained at higher energies. To combined this toy model with our measurements, if we aim to explain the spectral steepening above 10 TeV as a contribution from a Vela-type local source, the model must be able to reproduce both the H.E.S.S. measurements up to 40 TeV and the upper limits set by KM2A up to 160 TeV. This would require the local source to have an age of less than 2000 yrs, an electron cutoff energy extending beyond 10 TeV, or a diffusion coefficient several times lower than the typical value adapted in general models. However, no such nearby electron source has been identified so far. This "trilemma" to some extent disfavors the astrophysics source contribution hypothesis for high-energy CRE. Therefore, the upper limits of the CRE spectrum from KM2A can provide constraints on the existence of young astrophysical sources and the possible suppression of cosmic-ray diffusion at high energies.

Hadronic interaction models introduce significant uncertainties in estimating muon-poor events, even more than uncertainties in absolute flux normalization or cosmic-ray composition. Due to the insufficient discrimination power, the muon-free proton background overwhelms the electron event signals, allowing us to set only an upper limit on CRE events. The muon-poor events observed by KM2A constitute a small fraction of the total, limiting our ability to validate hadronic models. Furthermore, the limited coverage of KM2A's muon detector (~4%) reduces the statistics of the muon signal. The main challenge in distinguishing CREs from hadronic showers arises from proton-induced shower, particularly at lower energies. Secondary muons, produced by the decay of charged pions in hadronic interactions with the atmosphere, complicate the classification, as proton-induced EAS events can be misidentified as electronlike due to their muon-poor signatures. CRE flux measurements from IACTs show a similar level of dependence on hadronic interaction models or a preference for certain models [23,24], despite their different background estimation method, which fits simulated electron and proton background templates to observed data. To reduce model dependence, the deployment of a larger, more continuous muon detector with higher secondary muon counting accuracy is recommended. Such an enhancement could significantly improve cosmic-ray background rejection and enable precise CRE measurements above 25 TeV.

VI. CONCLUSION

In this study, we analyzed data collected by KM2A from July 2021 to July 2023 to set upper limits on the CRE flux in the energy range of 25 to 160 TeV. Our analysis reveals a strong dependence on hadronic interaction models during the estimation of the cosmic nuclei background. Currently, KM2A's background rejection capability is insufficient to mitigate this model dependence, allowing only upper limits on the CRE flux. Although KM2A's observations cannot rule out contributions from astrophysical sources at tens of TeV, they provide more stringent constraints at higher energies, particularly in the hundreds of TeV. These constraints can provide limits on younger astrophysical sources or smaller diffusion coefficients.

The issue of post-LHC hadronic interaction models remains a challenge, or equivalently, the muon-deficit problem, with the air-shower experiment community working to improve post-LHC high-energy hadronic interaction models. KM2A also contributes to this effort by measuring the attenuation length of muon content in air showers, which are sensitive indicators for validating these models. These ongoing efforts will enhance our ability to measure the highest energy part of the electron spectrum.

Given the rapidly falling flux at higher energies, space-based instruments will not be able to measure the CRE spectrum beyond certain thresholds. In contrast, ground-based arrays like KM2A and LHAASO will be crucial for continuing measurements in this energy range. To further enhance LHAASO's background rejection capability above 25 TeV, a novel approach utilizing WCDA as a continuous muon counter has been proposed, creating an additional muon-sensitive area. This innovation can improve KM2A's background rejection power, making it possible to measure the high-energy part of the electron spectrum in the future.

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DATA AVAILABILITY

The data that support the findings of this article are not publicly available. The data are available from the authors upon reasonable request.

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